

The φ Update in `mythesis.tex` versus `code/R/ar2`: A Step-by-Step Correspondence Audit

Technical Note — `thesis_phi_update_audit`

2026-08-01

Abstract

This note checks, step by step, whether the thesis description of how the AR coefficients φ enter and are updated in the MCMC matches `code/R/ar2`. The audit separates three questions that the phrase “the φ update” can mean, because the thesis answers them very differently. (A) How φ enters the Metropolis–Hastings update of the time-varying parameters — Appendix C of the thesis (`app:full-cond`); the code matches it exactly, including the head-of-series and end-of-series edge cases. (B) How φ itself is drawn — the code runs a component-wise random-walk Metropolis kernel with silent rejection on the stationarity triangle, targeting the *exact* stationary likelihood; the thesis does not describe this block at all. (C) The prior on φ — the thesis states `Unif(stationarity triangle)`, whereas the code uses independent $\mathcal{N}(0, 0.5^2)$ factors truncated to that triangle. (C) is a contradiction between text and code, (B) is an omission, (A) is correct. Two minor prose inaccuracies in Appendix C are also recorded. Section 6 gives paste-ready replacement text for `mythesis.tex`.

Contents

1	Scope, and a notation collision that must be cleared first	2
1.1	Two different objects are both called Y	2
1.2	Three distinct questions	2
2	The object φ parameterises	2
3	Layer (A): how φ enters the update of X_t^*	3
3.1	What the thesis says	3
3.2	Making the edge convention explicit	4
3.3	What the code does	4
3.4	Two prose inaccuracies in Appendix C	5
4	Layer (B): the φ block itself, which the thesis omits	5
4.1	(B1) The target	5
4.2	(B2) The likelihood is the exact stationary one	6
4.3	(B3) The kernel: component-wise random walk with silent rejection	6
4.4	(B4) Adaptation and bookkeeping	7
5	Layer (C): the prior, where text and code disagree	7
5.1	The two statements side by side	7
5.2	Why the difference is not cosmetic	7

1 Scope, and a notation collision that must be cleared first

1.1 Two different objects are both called Y

The thesis uses the symbol Y for two things.

- In the model, $Y_t(\mathbf{s})$ is the *response* at site $\mathbf{s}$ on day t (thesis `eq:fullmodel`), and $\mathbf{Y}_t$ is its n -vector over sites.
- In the thesis’s `sec:ar2` and throughout Appendix C (`app:full-cond`), $\{Y_t\}$ denotes “any scalar component of $\mathbf{w}_{tk}^*$, or z_{tk}^* or σ_{tk}^{2*} ” — that is, one *time-varying parameter on the transformed (standard-normal) scale*, at a fixed knot k .

Both meanings appear inside the same display, thesis `eq:full-cond`, whose likelihood factor is written $L(\mathbf{Y} | Y_t)$: bold $\mathbf{Y}$ is the data, scalar Y_t is the time-varying parameter. The companion note `tex/transform_ar_construction/` avoids the collision by writing X_t^* for the transformed time-varying parameter. **This note adopts X_t^* throughout**, reserving $\mathbf{Y}_t$ for the data.

Definition 1.1 (The three transformed channels). Fix a knot $k \in \{1, \dots, K\}$ and let $X_{1:n_t}^*$ denote any one of the scalar series

$$X_{tk}^* \in \{w_{tk}^{(1)*}, w_{tk}^{(2)*}, z_{tk}^*, \sigma_{tk}^{2*}\}, \quad (1)$$

obtained from the probability integral transforms `eq:wstar-eq:sigmastar` of the thesis. Each channel carries its own coefficient pair, $\varphi^{(w)}, \varphi^{(z)}, \varphi^{(\sigma)}$, shared across knots k (and, for w , across both coordinates $j = 1, 2$).

1.2 Three distinct questions

Once the notation is fixed, “does the thesis description of the φ update match the code?” splits into three questions, answered in Sections 3 to 5 and summarised in Table 1.

Table 1: Audit verdict. Thesis labels refer to `myLatex/thesis/mythesis.tex`; code paths are relative to `code/R/ar2/`.

	Question	Thesis	Verdict
(A)	How does φ enter the MH update of X_t^* ?	<code>app:full-cond</code> , <code>app:ar2-moments</code>	Matches , equation by equation (Section 3)
(B)	How is φ itself updated?	— (nothing)	Undocumented ; four features missing (Section 4)
(C)	What is the prior on φ ?	<code>eq:stage3-priors</code> : $\text{Unif}(S_2)$	Contradicts the code, which uses truncated $\mathcal{N}(0, 0.5^2)$ (Section 5)

2 The object φ parameterises

Definition 2.1 (Standardised stationary AR(2)). For each channel of Proposition 1.1, the prior is

$$X_t^* = \varphi_1 X_{t-1}^* + \varphi_2 X_{t-2}^* + \epsilon_t, \quad \epsilon_t \stackrel{\text{iid}}{\sim} \mathcal{N}(0, \sigma_\epsilon^2), \quad t \geq 3, \quad (2)$$

subject to the unit-variance normalisation $\gamma_0 := \text{Var}(X_t^*) = 1$, which the back-transform requires at *every* t , and to stationarity,

$$\varphi = (\varphi_1, \varphi_2) \in S_2 := \{|\varphi_2| < 1, \varphi_1 + \varphi_2 < 1, \varphi_2 - \varphi_1 < 1\}, \quad (3)$$

the open triangle with vertices $(-2, -1)$, $(2, -1)$, $(0, 1)$.

Proposition 2.2 (Yule–Walker moments). *Under Proposition 2.1,*

$$\gamma_0 = 1, \quad \gamma_1 = \frac{\varphi_1}{1 - \varphi_2}, \quad \gamma_2 = \varphi_1 \gamma_1 + \varphi_2, \quad \sigma_\epsilon^2 = 1 - \varphi_1 \gamma_1 - \varphi_2 \gamma_2, \quad (4)$$

all functions of φ alone.

Code. `get_ar2_standard_moments`, `auxfunctions_ar2.R` L9–54; the four assignments at L31–34 are (4) verbatim. **Thesis.** `eq:gamma12=eq:sigeps`, derived in Appendix B (`app:ar2-moments`). **Match.** Exact. The stationarity test (3) is `check_ar2_stability`, L2–7 (with a tolerance 10^{-8} on each strict inequality), matching thesis `eq:stab-triangle`.

Proposition 2.3 (Joint factorisation). *Under Proposition 2.1 the joint prior of $X_{1:n_t}^*$ factorises as $p(X_{1:n_t}^*) = p_1(X_1^*) p_2(X_2^* | X_1^*) \prod_{s=3}^{n_t} p_s(X_s^* | X_{s-1}^*, X_{s-2}^*)$ with*

$$p_1(x) = \mathcal{N}(x; 0, 1), \quad (5)$$

$$p_2(x | x_1) = \mathcal{N}(x; \gamma_1 x_1, 1 - \gamma_1^2), \quad (6)$$

$$p_s(x | x_{s-1}, x_{s-2}) = \mathcal{N}(x; \varphi_1 x_{s-1} + \varphi_2 x_{s-2}, \sigma_\epsilon^2), \quad s \geq 3. \quad (7)$$

Code. `get_ar2_conditional_params`, `auxfunctions_ar2.R` L56–92, whose three branches are exactly (5) (L66–68), (6) (L70–85), (7) (L87–91). **Thesis.** `eq:joint-prior` together with the initial pair `eq:joint-init`; (6) is the conditional of that bivariate normal. **Match.** Exact.

Remark 2.4 (φ enters through two channels, not one). This is the structural fact behind everything that follows. In (7), φ enters through the mean coefficients *and* through σ_ϵ^2 . In (6) it enters only through $\gamma_1 = \varphi_1/(1 - \varphi_2)$. The $t = 2$ factor is therefore φ -dependent, and

$$\mathcal{N}(\gamma_1 x_1, 1 - \gamma_1^2) = \mathcal{N}(\varphi_1 x_1, \sigma_\epsilon^2) \iff \varphi_2 = 0. \quad (8)$$

Using the right-hand form (the recursion extrapolated backwards with a fictitious $X_0^* = 0$) breaks $\text{Var}(X_2^*) = 1$ and hence breaks the target marginal after the back-transform. The code uses the left-hand form (“Fix 1”, `auxfunctions_ar2.R` L74–84). Only p_1 is φ -free.

3 Layer (A): how φ enters the update of X_t^*

3.1 What the thesis says

Appendix C states that a proposal for X_t^* changes three factors of Proposition 2.3 — the self factor, the lag-1 factor and the lag-2 factor — so that

$$\pi(X_t^* | X_{-t}^*, \mathbf{Y}) \propto \underbrace{L(\mathbf{Y} | X_t^*)}_{\text{spatial likelihood}} \cdot \underbrace{p_t(X_t^* | X_{t-1}^*, X_{t-2}^*)}_{\text{self}} \cdot \underbrace{p_{t+1}(X_{t+1}^* | X_t^*, X_{t-1}^*)}_{\text{lag-1}} \cdot \underbrace{p_{t+2}(X_{t+2}^* | X_{t+1}^*, X_t^*)}_{\text{lag-2}}, \quad (9)$$

with the convention that factors referencing a non-existent time index are dropped, and that under a symmetric random-walk proposal $X_t^{*(c)} \sim \mathcal{N}(X_t^*, s^2)$ the acceptance probability is $R = \min\{1, \pi(X_t^{*(c)} | \cdot) / \pi(X_t^* | \cdot)\}$.

3.2 Making the edge convention explicit

Write $\mathcal{S}(t) \subseteq \{1, \dots, n_t\}$ for the set of factor indices in which X_t^* appears. Proposition 2.3 gives

$$\mathcal{S}(t) = \{t, t+1, t+2\} \cap \{1, \dots, n_t\}, \quad (10)$$

which unpacks as in Table 2. The acceptance ratio actually evaluated is then

$$\log R = \underbrace{\log L(\mathbf{Y} | X_t^{*(c)}) - \log L(\mathbf{Y} | X_t^*)}_{\text{likelihood part}} + \sum_{s \in \mathcal{S}(t)} \left[\log p_s(\cdot | \cdot) \Big|_{X_t^* = X_t^{*(c)}} - \log p_s(\cdot | \cdot) \Big|_{X_t^*} \right]. \quad (11)$$

Table 2: The factor set $\mathcal{S}(t)$ of (10), written out. The $t = 1$ and $t = 2$ rows are the “initial-pair” cases: since $p(X_1^*, X_2^*) = p_1(X_1^*) p_2(X_2^* | X_1^*)$, the head of the series needs no special code path.

t	Factors containing X_t^*	Code guard
1	$p_1(X_1^*), p_2(X_2^* X_1^*), p_3(X_3^* X_2^*, X_1^*)$	self ($t=1$ branch), $\mathbf{t} < \mathbf{nt}$, $\mathbf{t}+2 \leq \mathbf{nt}$
2	$p_2(X_2^* X_1^*), p_3(\cdot), p_4(X_4^* X_3^*, X_2^*)$	self ($t=2$ branch), $\mathbf{t} < \mathbf{nt}$, $\mathbf{t}+2 \leq \mathbf{nt}$
$3 \leq t \leq n_t - 2$	$p_t(\cdot), p_{t+1}(\cdot), p_{t+2}(\cdot)$	all three
$n_t - 1$	$p_{n_t-1}(\cdot), p_{n_t}(\cdot)$	$\mathbf{t}+2 \leq \mathbf{nt}$ false
n_t	$p_{n_t}(\cdot)$	$\mathbf{t} < \mathbf{nt}$ false

3.3 What the code does

All three updaters implement (11) with the guards of Table 2; see Table 3. The lag-2 term is the “Fix 2” patch documented in `tex/where_we_are/`, whose Proposition on detailed balance shows that omitting it makes the chain target a modified stationary law whose lag-2 autocorrelation misses γ_2 .

Table 3: Where each factor of (9) lives, by channel. All files in `code/R/ar2/`.

Channel	Updater (file <code>update_params_ar2.R</code>)	self	lag-1	lag-2
$\sigma^{2*}, K = 1$	<code>updateTauTS_AR2</code>	L51–59	L62–75	L82–95
$\sigma^{2*}, K > 1$	<code>updateTauTS_AR2</code>	L142–150	L153–166	L172–185
z^*	<code>updateZTS_AR2</code>	L392–400	L403–416	L421–434
w^*	<code>updateKnotsTS_AR2</code>	L548–555	L566–579	L584–597

Proposition 3.1 (Layer (A) verdict). *For every channel and every t , the set of prior factors entered into the acceptance ratio by the code equals $\mathcal{S}(t)$ of (10), and each factor is evaluated with the branch of Proposition 2.3 matching its own time index. The code therefore implements (9) exactly as Appendix C describes it.*

Verification. Take `updateZTS_AR2` as representative. The self term (L392–400) calls `ar2_conditional_log_density` with time index t and lags

```
lag1 <- if (t >= 2) z.star[k, t - 1] else NULL
lag2 <- if (t >= 3) z.star[k, t - 2] else NULL
```

so that by Proposition 2.3 this selects (5) at $t = 1$, (6) at $t = 2$, (7) otherwise. The lag-1 term (L403–416) is guarded by `t < nt` and passes time index $t + 1$ with the candidate as its *lag-1* argument and `z.star[k, t-1]` as its lag-2 argument; at $t = 1$ that argument is `NULL` and the $t+1 = 2$ branch (6) is selected, which is why the head of the series needs no special case. The lag-2 term (L421–434) is guarded by `t+2 <= nt` and passes time index $t + 2$ with `z.star[k, t+1]` as lag-1 and the candidate as lag-2. Reading off the three time indices gives exactly the rows of Table 2. The same three blocks appear at the line numbers of Table 3 for the other channels. $\square$

3.4 Two prose inaccuracies in Appendix C

- (A1) **The proposal is not scalar for w^* .** Appendix C writes $X_t^{*(c)} \sim \mathcal{N}(X_t^*, s^2)$, a single-component move. That is what `updateTauTS_AR2` and `updateZTS_AR2` do (one (k, t) cell at a time). `updateKnotsTS_AR2`, however, proposes all $K \times 2$ knot coordinates of day t jointly and accepts or rejects them as one block (`update_params_ar2.R` L523–526, single `log(runif(1)) < R` at L599–606):

$$w_{tk}^{*(c)} = w_{tk}^* + s_{kt} \varepsilon_k, \quad \varepsilon_k \stackrel{\text{iid}}{\sim} \mathcal{N}(\mathbf{0}, \mathbf{I}_2), \quad k = 1, \dots, K, \quad (12)$$

which is still a symmetric random walk, so (11) is unchanged in form, but it is a block move and the text should say so.

- (A2) **The adaptation band is $[0.25, 0.50]$, not $0.3\text{--}0.4$.** `mhupdate` (`auxfunctions.R` L99–113) rescales s by 0.8 when the running acceptance rate falls below 0.25 and by 1.2 when it exceeds 0.50, every 50 attempts, and this is applied only while `iter < burn/2` (`mcmc_ar2.R` L556). The stopping of adaptation before the sampling phase is worth stating, since it is what makes the adaptive scheme valid.

4 Layer (B): the φ block itself, which the thesis omits

Appendix C ends at (9). Nowhere does the thesis say how $\varphi^{(w)}, \varphi^{(z)}, \varphi^{(\sigma)}$ are drawn. The code does it in `updatePhiAR2TS` (`update_params_ar2.R` L225–340), called once per sweep from each channel’s updater (L201–205, L449–453, L613–617). Four features need documenting.

4.1 (B1) The target

Conditionally on the transformed latents, φ is independent of the data and of all other parameters, so its full conditional involves only the AR prior and its own prior:

$$\pi(\varphi | X^*) \propto \left[\prod_{k=1}^K p(X_{1:n_t, k}^* | \varphi) \right] p(\varphi) \mathbf{1}\{\varphi \in S_2\}, \quad (13)$$

the product running over knots (and, for w^* , over both coordinates), which are independent replicates of the same AR(2) law.

4.2 (B2) The likelihood is the exact stationary one

Expanding $p(X_{1:n_t,k}^* | \varphi)$ with Proposition 2.3 and dropping the φ -free factor p_1 , the log-likelihood used in the acceptance ratio is

$$\ell(\varphi) = \underbrace{\sum_k \log \mathcal{N}(X_{2k}^*; \gamma_1 X_{1k}^*, 1 - \gamma_1^2)}_{\text{initial block, } \varphi\text{-dependent}} + \underbrace{\sum_k \sum_{t=3}^{n_t} \log \mathcal{N}(X_{tk}^*; \varphi_1 X_{t-1,k}^* + \varphi_2 X_{t-2,k}^*, \sigma_\epsilon^2)}_{\text{transitions (Box-Jenkins conditional likelihood)}}. \quad (14)$$

Code. `ar2_stationary_loglik`, `auxfunctions_ar2.R` L122–133; the two summands are L131 and L132 respectively.

Remark 4.1 (Why the initial block cannot be dropped, and why AR(1) is exempt). Keeping only the transition sum is the Box–Jenkins conditional likelihood. For AR(1) that is exact up to a φ -free constant, because the only discarded factor is $p_1 = \mathcal{N}(0, 1)$, which does not involve φ . For AR(2) the discarded factor is $p_1 \cdot p_2$, and by Proposition 2.4 p_2 *does* involve φ through γ_1 . Dropping it makes the φ block and the X^* block target different joint laws, i.e. the two Gibbs blocks become incompatible; see `tex/ar2_phi_exact_likelihood/`. Appendix C already argues the dual statement for the X^* block (“all three prior factors must appear”); the same γ_1 dependence produces this second trap, and the thesis states only the first half.

4.3 (B3) The kernel: component-wise random walk with silent rejection

One sweep updates φ_1 and then φ_2 , the second step conditioning on the just-updated first. For φ_1 (the φ_2 step is identical with the roles exchanged):

$$\varphi_1^{(c)} \sim \mathcal{N}(\varphi_1, s_1^2) \quad (\text{untruncated}), \quad (15)$$

$$\log R = \ell(\varphi_1^{(c)}, \varphi_2) - \ell(\varphi_1, \varphi_2) + \log p(\varphi_1^{(c)}) - \log p(\varphi_1), \quad (16)$$

$$\varphi_1^{\text{new}} = \begin{cases} \varphi_1^{(c)}, & \text{if } (\varphi_1^{(c)}, \varphi_2) \in S_2 \text{ and } \log U < \log R, \\ \varphi_1, & \text{otherwise (in particular, whenever the candidate leaves } S_2). \end{cases} \quad U \sim \text{Unif}(0, 1), \quad (17)$$

Code. L281–305 (φ_1) and L308–331 (φ_2); the stationarity test at L283/L310 is the $\mathbf{1}\{\cdot \in S_2\}$ of (17); the moment recomputation is wrapped in `tryCatch` so that a numerically degenerate candidate is also silently rejected.

Lemma 4.2 (Silent rejection needs no Hastings correction). *Let q be symmetric, $q(\varphi' | \varphi) = q(\varphi | \varphi')$, and let the kernel accept only candidates in S_2 , with probability $\alpha(\varphi, \varphi') = \min\{1, \pi(\varphi')/\pi(\varphi)\}$, otherwise remaining at φ . Then for $\varphi, \varphi' \in S_2$,*

$$\pi(\varphi) q(\varphi' | \varphi) \alpha(\varphi, \varphi') = \min\{\pi(\varphi), \pi(\varphi')\} q(\varphi' | \varphi) = \pi(\varphi') q(\varphi | \varphi') \alpha(\varphi', \varphi), \quad (18)$$

so detailed balance holds with respect to π restricted to S_2 ; mass proposed outside S_2 contributes only to the holding probability. No Jacobian or truncation ratio enters.

Remark 4.3 (Do not “fix” this to match the AR(1) code). The AR(1) reference updater `updatePhiTS` (`update_params.R` L1156–1202) instead draws from a normal truncated to $(-1, 1)$ and carries the corresponding asymmetry correction, terms `curlog.cand` – `canlog.cand` at L1192. The audit in `tex/where_we_are/` records that an attempt to impose the analogous correction on the AR(2) silent-rejection kernel biased the chain toward the boundary of S_2 and was reverted. Both kernels are valid; they are simply different, and the thesis should describe the one that is used.

4.4 (B4) Adaptation and bookkeeping

The proposal scales s_1, s_2 are tuned by the same `mhupdate` rule as in Section 3.4, separately for each channel (`mcmc_ar2.R` L567–581 for $\mathbf{w}$, and the matching blocks for σ^2 and z), and only while `iter < burn/2`. Attempt counters are incremented before the stationarity test (L229–230), so an out-of-triangle proposal counts as a rejection — which is the correct accounting for the kernel of Proposition 4.2, since such a move genuinely fails.

5 Layer (C): the prior, where text and code disagree

5.1 The two statements side by side

The thesis, `eq:stage3-priors`, states

$$(\varphi_1^{(\cdot)}, \varphi_2^{(\cdot)}) \sim \text{Unif}(S_2), \quad \text{i.e.} \quad p_{\text{thesis}}(\boldsymbol{\varphi}) \propto \mathbf{1}\{\boldsymbol{\varphi} \in S_2\}. \quad (19)$$

The code, through the defaults `prior.mean = 0`, `prior.sd = 0.5` of `updatePhiAR2TS` (L228) used at L295–297 and L322–324, implements

$$p_{\text{code}}(\boldsymbol{\varphi}) \propto \mathcal{N}(\varphi_1; 0, 0.5^2) \mathcal{N}(\varphi_2; 0, 0.5^2) \mathbf{1}\{\boldsymbol{\varphi} \in S_2\}, \quad (20)$$

the indicator being realised implicitly by (17). The arguments `prior.mean` and `prior.sd` are never supplied by any production caller — `updateTauTS_AR2`, `updateZTS_AR2`, `updateKnotsTS_AR2` all call with the defaults — and are overridden only in `tests/test_fix3_phi_hastings.R`. So (20) is effectively hardcoded. The AR(1) baseline is the same story one dimension down: $\mathcal{N}(0, 0.5^2)$ truncated to $(-1, 1)$ (`update_params.R` L1190–1191).

5.2 Why the difference is not cosmetic

Step 1: the penalty is not flat. On S_2 ,

$$\log p_{\text{code}}(\boldsymbol{\varphi}) - \log p_{\text{code}}(\mathbf{0}) = -\frac{\varphi_1^2 + \varphi_2^2}{2 \cdot 0.5^2} = -2(\varphi_1^2 + \varphi_2^2), \quad (21)$$

whereas (19) makes this identically 0. At the generating values of the simulation study the penalty is 1.53 nats at $(\varphi_1, \varphi_2) = (0.80, -0.35)$ and 1.33 nats at $(0.15, 0.80)$, relative to the origin. These are small against a likelihood built from $K(n_t - 1)$ Gaussian factors, which is why `tex/ar2_phi1_prior_support/` measures a posterior shrinkage of at most 0.04 and concludes “no code change”. The point is not that the sampler is wrong — it validly targets the posterior induced by (20) — but that the thesis reports a prior the study did not use.

Step 2: the scale is calibrated to the wrong support. Under S_2 the marginal supports are

$$\varphi_1 \in (-2, 2), \quad \varphi_2 \in (-1, 1), \quad (22)$$

so φ_1 has twice the range of the AR(1) coefficient while (20) keeps the AR(1) scale 0.5. For the AR(1) pairing the match is deliberate, $\Phi(1/0.5) - \Phi(-1/0.5) \approx 0.954$, about 95% of the untruncated mass inside the support. For φ_1 under S_2 the same scale leaves $\Phi(4) - \Phi(-4) > 0.9999$ inside, i.e. the prior is far more informative relative to its support than the AR(1) prior it was ported from. The mismatch is confined to φ_1 ; for φ_2 support and scale still match.

Step 3: the two orders are not prior-nested. Setting $\varphi_2 = 0$ in (20) recovers the AR(1) prior on that section. But the AR(2) fit never conditions on $\varphi_2 = 0$; the comparison in the thesis is against the *marginal* induced on φ_1 ,

$$p_{\text{code}}(\varphi_1) \propto \mathcal{N}(\varphi_1; 0, 0.5^2) \int_{\{\varphi_2: (\varphi_1, \varphi_2) \in S_2\}} \mathcal{N}(\varphi_2; 0, 0.5^2) d\varphi_2 \neq p_{\text{AR}(1)}(\varphi_1), \quad (23)$$

because the inner integral depends on φ_1 through the triangle's φ_1 -section. Every AR(2)-versus-AR(1) contrast in the thesis (`tab:ar2-sim-paired`, `tab:timeblock-brier`, `tab:ozone-brier-ci`) therefore carries a small prior confound alongside the change in order. Recording this caveat in the paper is one of the two follow-ups left open by `tex/ar2_phi1_prior_support/` (§Status and decision, 2026-07-15).

6 Recommended edits to mythesis.tex

Nothing in `code/R/ar2` needs to change; all three edits are textual.

E1 — correct the prior (eq:stage3-priors). Replace the φ entry by the truncated-normal form and add the non-nestedness caveat where AR(1) and AR(2) are compared:

```
(\phi~{\(\cdot\)}_1,\, \phi~{\(\cdot\)}_2)
&\mathrlap{\,;\sim\,}; N(0,\, \sigma^2_\phi \,bI_2)\ \text{\texttt{\text{restricted to the}}}
\text{\texttt{stationarity triangle \ref{eq:stab-triangle}}},\}
```

% ... and in the hyperparameter sentence: $\sigma_\phi = 0.5$.

% Caveat sentence, to sit with the AR(1)-vs-AR(2) comparison:

% Because the triangle \ref{eq:stab-triangle} gives ϕ_1 the support

% $(-2,2)$ while ϕ_2 retains $(-1,1)$, the common scale

% $\sigma_\phi = 0.5$ is calibrated to the AR(1) support; the induced

% marginal prior on ϕ_1 under the AR(2) model is therefore not the

% AR(1) prior, and the AR(2)-versus-AR(1) contrasts carry a small prior

% component alongside the change in order. The posterior shrinkage this

% induces was measured at no more than 0.04 across the generating values

% used here.

E2 — add the φ block to Appendix C. A subsection after the thesis’s `eq:mh-ratio` covering, in order: the target (13); the exact stationary likelihood (14) with Proposition 4.1 on why the initial block survives for AR(2) but is vacuous for AR(1); the component-wise scan with silent rejection (15)–(17) and Proposition 4.2; and one sentence that a single pair $\varphi^{(\cdot)}$ is shared across knots (and across both coordinates of $\mathbf{w}^*$).

E3 — two corrections inside Appendix C. Change “an acceptance rate near 0.3–0.4” to the actual rule, adaptation by factors 0.8/1.2 whenever the running rate leaves $[0.25, 0.50]$, stopped after the first half of burn-in; and note that for $\mathbf{w}^*$ the day- t proposal (12) is a block move over all $K \times 2$ coordinates rather than the scalar move written for the other two channels.

E4 — optional, but it is the cause of the confusion. Rename the appendix’s time-varying parameter from Y_t to X_t^* (or U_t), so that (9) does not carry $\mathbf{Y}$ and Y_t with two different meanings in one display. This matches `tex/transform_ar_construction/`.

7 Summary

- (A) The description of how φ enters the update of the time-varying parameters — Appendices B and C — **matches the code exactly**, including the Yule–Walker standardisation (4), the stationary $t = 2$ conditional (6), and the three-factor acceptance ratio (11) with its edge conventions. Two prose details need correcting (Section 3.4).
- (B) The update of φ *itself* is **not described anywhere in the thesis**, although it embodies two deliberate and non-obvious decisions: the exact stationary likelihood (14) and the silent-rejection kernel (17).
- (C) The one place the thesis does speak about φ directly, its prior, **contradicts the code**: $\text{Unif}(S_2)$ in the text (19) versus truncated independent $\mathcal{N}(0, 0.5^2)$ in the implementation (20), with a documented consequence for the AR(1)-versus-AR(2) comparisons (23).